

Transfer functions

Poles, Zeros, Step response, and Simple Feedback Loops.

Notation:

- $r(t)$ reference/input
- $y(t)$ output
- $v(t)$ variable/controlled output
- $d(t)$ disturbance
- $u(t)$ step function

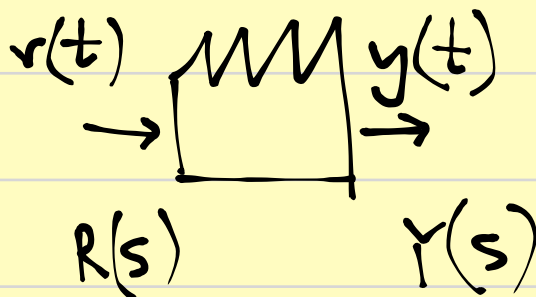
Transfer function (TF)

$\frac{2}{3} = 0'6$ How many times 2 is contained in 3.

A TF in the "s" domain $G(s) = \frac{Y(s)}{R(s)}$ for zero initial conditions.
How many times the output is contained in the input.



First order model.



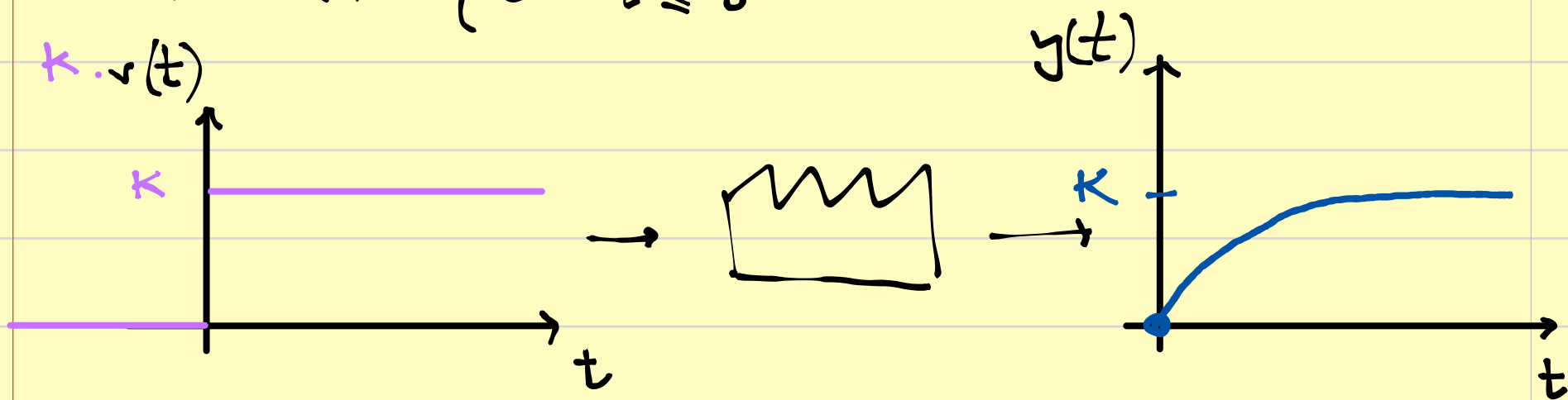
$$\tau \dot{y}(t) + y(t) = K r(t)$$

$$y(0) = 0$$

Laplace: $\tau s Y(s) + Y(s) = K R(s)$

$$Y(s) [\tau s + 1] = K R(s) \rightarrow G(s) = \frac{Y(s)}{R(s)} = \frac{K}{1 + \tau s}$$

$$r(t) = u(t) = \begin{cases} 1 & t > 0 \\ 0 & t \leq 0 \end{cases}$$



$$G(s) = \frac{Y(s)}{R(s)} = \frac{K}{1 + \tau s} \rightarrow Y(s) = R(s) \cdot \frac{K}{1 + \tau s} = \frac{1}{s} \cdot \frac{K}{1 + \tau s}$$

↑
Laplace Step Funct.

$$Y(s) = \frac{K}{\tau} \cdot \frac{1}{s} \cdot \frac{1}{s + \frac{1}{\tau}} = \frac{A}{s} + \frac{B}{s + \frac{1}{\tau}}$$

$$\frac{K}{\tau} = A \left(s + \frac{1}{\tau} \right) + B \cdot s$$

$$s = 0 \rightarrow \frac{K}{\tau} = A \cdot \frac{1}{\tau} \rightarrow A = K$$

$$s = -\frac{1}{\tau} \rightarrow \frac{K}{\tau} = -\frac{B}{\tau} \rightarrow B = -K$$

$$Y(s) = K \left[\frac{1}{s} - \frac{1}{s + \frac{1}{\tau}} \right] \xrightarrow{\mathcal{L}^{-1}}$$

$$y(t) = K \left[1 - e^{-t/\tau} \right]$$

K : static gain τ : time constant

A general TF has the form $G(s) = \frac{N(s)}{D(s)}$

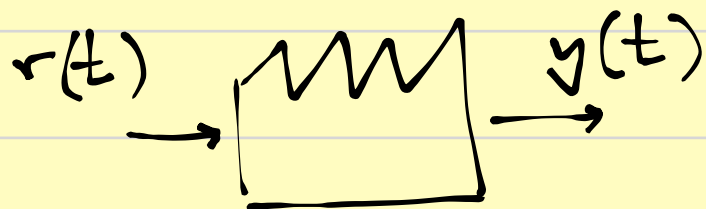
- The ZEROS are the roots of $N(s)$. $N(s) = 0$
- The POLES are the roots of $D(s)$. $D(s) = 0$
- The ORDER of the system is the degree of the denominator $D(s)$.

Stability rule for linear systems

If all POLES satisfy $\text{Re}(s) < 0$ then the linear system is stable
If one POLE has $\text{Re}(s) > 0$ then the system is unstable.

- POLES determine the natural dynamics.
- ZEROS shape how the input enters the output.

Second order model.

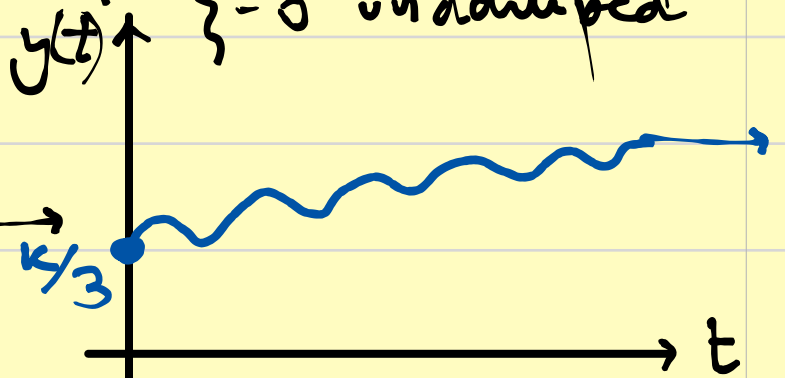
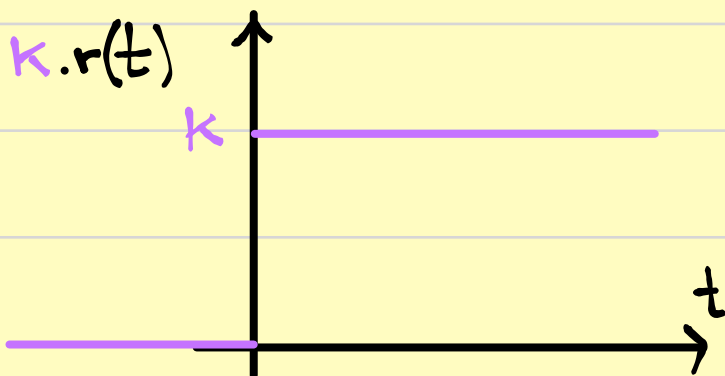


$$\ddot{y}(t) + 2\omega_n \zeta \dot{y}(t) + \omega_n^2 y(t) = \omega_n^2 r(t)$$

$$G(s) = \frac{Y(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta \omega_n s + \omega_n^2}$$

The damping classification appears:

$$\begin{cases} \zeta > 1 & \text{overdamped} \\ \zeta = 1 & \text{critically damped} \\ 0 < \zeta < 1 & \text{underdamped} \\ \zeta = 0 & \text{undamped} \end{cases}$$



$$Y(s) = R(s) \cdot \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{k}{s} \cdot \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Example: $\omega_n = 2$ $2\zeta\omega_n = 2 \rightarrow \zeta = \frac{1}{2}$

$$Y(s) = \frac{k}{s} \cdot \frac{4}{s^2 + 2s + 4} = 4k \cdot \frac{1}{s} \cdot \frac{1}{(s - (-1 + j\sqrt{3})) (s - (-1 - j\sqrt{3}))}$$

$$s^2 + 2s + 4 = 0 \rightarrow s = \frac{-2 \pm \sqrt{4 - 16}}{2} = -1 \pm j\sqrt{3}$$

$$= \frac{A}{s} + \frac{B}{s - (-1 + j\sqrt{3})} + \frac{C}{s - (-1 - j\sqrt{3})}$$

$$4k = A \left[(s - (-1 + j\sqrt{3})) (s - (-1 - j\sqrt{3})) \right] + B \cdot s (s - (-1 - j\sqrt{3})) + C \cdot s (s - (-1 + j\sqrt{3}))$$

$$s = 0 \rightarrow 4k = A \cdot [(1 - j\sqrt{3})(1 + j\sqrt{3})] = A \cdot [1^2 + (\sqrt{3})^2] = A \cdot 4$$

$$A = k$$

$$s = -1 - j\sqrt{3} \rightarrow 4k = C \cdot [-1 - j\sqrt{3}] \left[-1 - j\sqrt{3} - (-1 + j\sqrt{3}) \right] =$$

$$= C \cdot [-1 - j\sqrt{3}] \cdot 3$$

$$C = \frac{4}{3} k \cdot \frac{1}{-1 - j\sqrt{3}} \cdot \frac{-1 + j\sqrt{3}}{-1 + j\sqrt{3}} =$$

$$= \frac{4}{3} k \cdot \frac{-1 + j\sqrt{3}}{4}$$

$$c = \frac{k}{3}(-1+j\sqrt{3})$$

$$s = -1+j\sqrt{3} \rightarrow 4k = B \cdot [-1+j\sqrt{3}] \cdot [-1+j\sqrt{3} - (-1-j\sqrt{3})]$$

$$= B \cdot [-1+j\sqrt{3}] \cdot 3$$

$$B = \frac{4k}{3} \cdot \frac{1}{-1+j\sqrt{3}} \cdot \frac{-1-j\sqrt{3}}{-1-j\sqrt{3}} =$$

$$= \frac{4k}{3} \cdot \frac{-1-j\sqrt{3}}{4}$$

$$B = \frac{k}{3} \cdot (-1-j\sqrt{3})$$

$$Y(s) = \frac{k}{s} + \frac{k}{3}(-1-j\sqrt{3}) \cdot \frac{1}{s-(-1+j\sqrt{3})} + \frac{k}{3}(-1+j\sqrt{3}) \cdot \frac{1}{s-(-1-j\sqrt{3})}$$

$\mathcal{L}^{-1}$

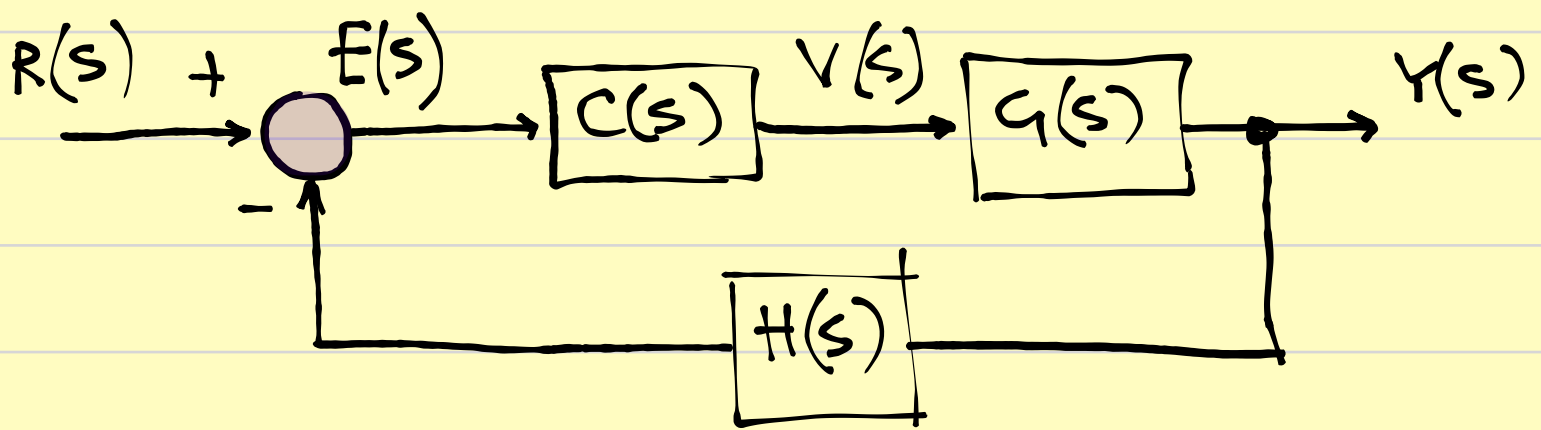
$$y(t) = k + \frac{k}{3}(-1-j\sqrt{3}) e^{(-1+j\sqrt{3})t} + \frac{k}{3}(-1+j\sqrt{3}) e^{(-1-j\sqrt{3})t}$$

$$y(0) = k + \frac{k}{3}(-1-j\sqrt{3}) + \frac{k}{3}(-1+j\sqrt{3})$$

$$= k + \frac{-k}{3} - \frac{k}{3} = \frac{k}{3}$$

$$y(t \rightarrow \infty) = k$$

Simple Feedback Loops



Error: $E(s) = R(s) - Y(s) \cdot H(s)$ Error $\equiv E(s)$

$$V(s) = E(s) \cdot C(s)$$

Output of Controller $\equiv V(s)$

$$Y(s) = G(s) \cdot V(s)$$

Output of the Plant $\equiv Y(s)$

Substitute: $Y(s) = G(s) \cdot E(s) \cdot C(s) =$

$$= G(s) \cdot C(s) \cdot [R(s) - Y(s) \cdot H(s)]$$

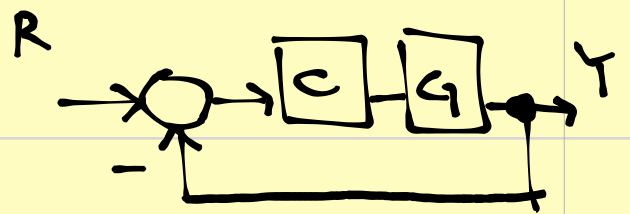
$$= G(s) \cdot C(s) \cdot R(s) - G(s) \cdot C(s) \cdot H(s) \cdot Y(s)$$

$$Y(s) [1 + G(s) \cdot C(s) \cdot H(s)] = G(s) \cdot C(s) \cdot R(s)$$

$$\frac{Y(s)}{R(s)} = \frac{G(s) \cdot C(s)}{1 + G(s) \cdot C(s) \cdot H(s)}$$

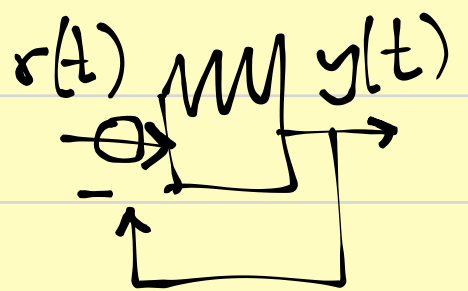
$$H(s) = 1$$

$$\frac{Y(s)}{R(s)} = \frac{G(s)C(s)}{1 + G(s)C(s)}$$



Closed-loop example: inventory/throughput regulation

$r(t)$: desired throughput (demand)
 $y(t)$: actual throughput

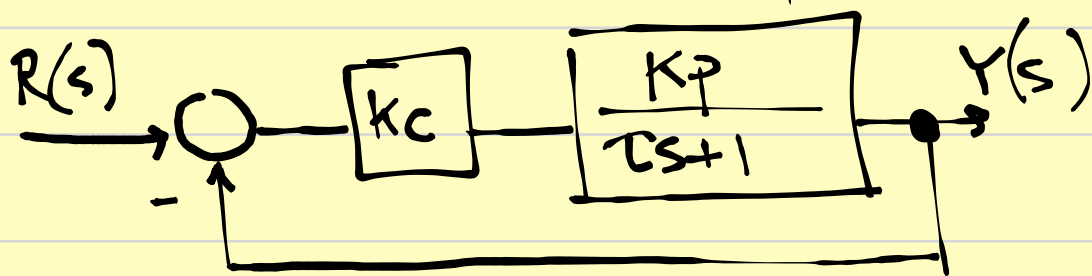


The plant is first order: $G(s) = \frac{k_p}{\tau s + 1}$

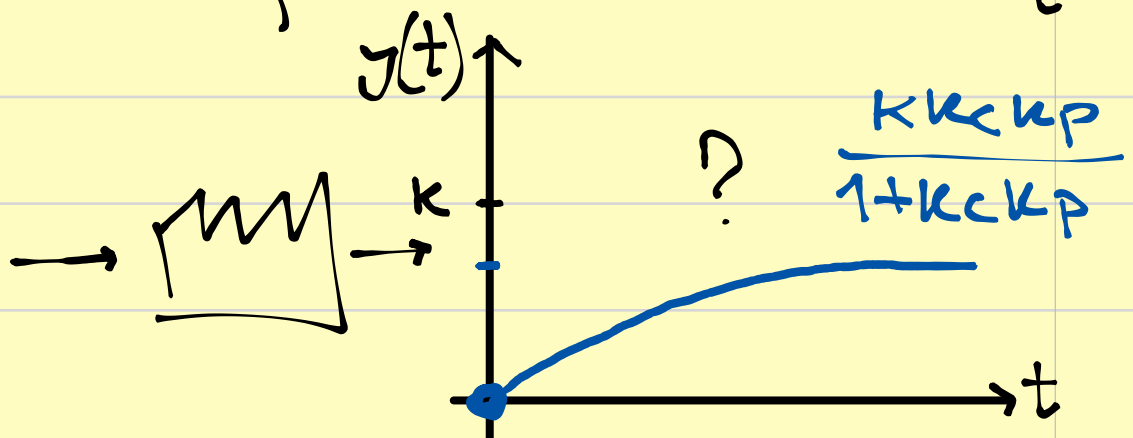
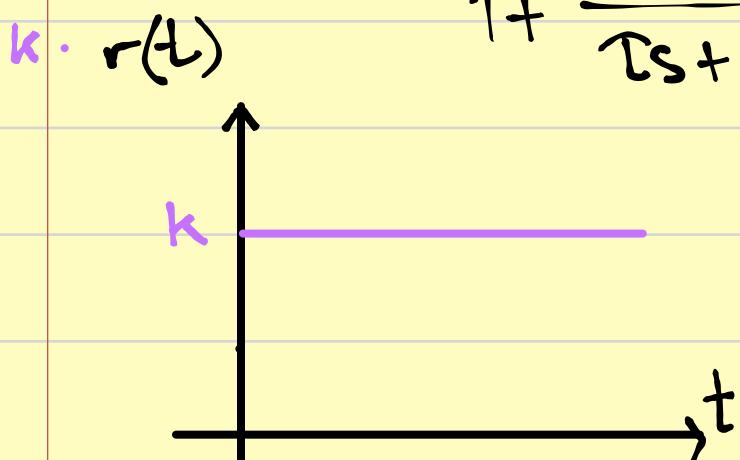
The controller is proportional: $C(s) = K_c$

Assume unity feedback

$$T(s) = \frac{Y(s)}{R(s)} = \frac{K_c \cdot K_p}{\tau s + 1 + K_c K_p}$$



$$T(s) = \frac{\frac{K_c K_p}{\tau s + 1}}{1 + \frac{K_c K_p}{\tau s + 1}} = \frac{K_c K_p}{K_c K_p + 1 + \tau s} = \frac{K_c K_p}{\tau} \cdot \frac{1}{s + \frac{1 + K_c K_p}{\tau}}$$



$$Y(s) = R(s) \cdot T(s) = \frac{k}{s} \cdot \frac{K_c K_p}{\tau} \cdot \frac{1}{s + \frac{1 + K_c K_p}{\tau}}$$

$$= \frac{k K_c K_p}{\tau} \cdot \frac{1}{s} \cdot \frac{1}{s + \frac{1 + K_c K_p}{\tau}} = \frac{A}{s} + \frac{B}{s + \frac{1 + K_c K_p}{\tau}}$$

$$\frac{k K_c K_p}{\tau} = A \left(s + \frac{1 + K_c K_p}{\tau} \right) + B \cdot s$$

$$s = 0 \rightarrow \frac{k K_c K_p}{\tau} = A \cdot \frac{1 + K_c K_p}{\tau} \rightarrow A = \frac{k K_c K_p}{1 + K_c K_p}$$

$$s = -\frac{1 + K_c K_p}{\tau} \rightarrow \frac{k K_c K_p}{\tau} = -B \frac{1 + K_c K_p}{\tau} \rightarrow B = \frac{-k K_c K_p}{1 + K_c K_p}$$

$$Y(s) = \frac{k K_c K_p}{1 + K_c K_p} \left[\frac{1}{s} - \frac{1}{s + \frac{1 + K_c K_p}{\tau}} \right]$$

$$\mathcal{L}^{-1} \left\{ y(t) = \frac{k K_c K_p}{1 + K_c K_p} \left[1 - e^{-t \cdot \frac{1 + K_c K_p}{\tau}} \right] \right\}$$

First order
Closed loop

$$y(t) = k \cdot \left[1 - e^{-t/\tau} \right]$$

First order
Open loop

This is useful for :

- Inventory Regulation
- Release Rate "
- Backlog Reduction
- Conveyor Control
- Warehouse Climate Control
- Oscillations Supply Chains.
(Bullwhip)

